

PRIMITIVE ROOTS

$$\langle a \rangle = G$$

Definition: $\text{ord}_n a$ is the smallest number m such that $a^m \equiv 1 \pmod n$.

Important fact: $\text{ord } a \mid \varphi(n)$

Definition: We say r is a primitive root modulo n if $\text{ord}_n r = \varphi(n)$.

Set of invertible units mod n

note: $\{1, r, r^2, \dots, r^{\varphi(n)-1}\} = U_n$

example: $n = 13$ $\varphi(13) = 12$

$\Rightarrow$ possible orders of elements are $\cancel{X} 2, 3, 4, 6, 12$

$$2^1 = 2, 2^2 = 4, 2^3 = 8, 2^4 = 16 = 3, 2^6 = 12$$

$\Rightarrow$ order of 2 must be 12 $\Rightarrow$ 2 is a primitive root.

$$n = 8 \quad \varphi(8) = 2^3 - 2^2 = 4$$

possible orders 1, 2, 4; possible primitive roots

$$\cancel{X} 3, 5, 7$$

$$3^2 = 9 = 1, 5^2 = 1, 7^2 = 1$$

$\Rightarrow$ no primitive roots modulo 8.

Lagrange's Theorem

For a finite group G and any subgroup H of G the order of H , $|H|$, divides the order of G , $|G|$. In other words

$$|G| = [G:H] \cdot |H|$$

where $[G:H]$ is the index of H in G (the number of distinct left cosets of H in G).

proof: Define the left coset of H in G for any $g \in G$ as $gH := \{gh \mid h \in H\}$.

Facts: Two cosets of G are either identical or disjoint.

Suppose $\exists x$ with $x = gh = g'h'$ and show $gH \subseteq g'H$ & $g'H \subseteq gH$.

Every element of G belongs to some coset gH .
The set of all left cosets of H forms a partition of G : $G = \bigcup gH$

Now for any $g \in G$, the map $f: H \rightarrow gH$ defined by $f(h) = gh$ is a bijection. Because if $gh_1 = gh_2$, then multiplying by g^{-1} yields $h_1 = h_2$, hence injective. Also, every element in gH is of the form gh for some $h \in H$, hence f is surjective; $\Rightarrow$ bijective.

Therefore, each coset gH has the same number of elements as H , $|gH| = |H|$.

Let k be the number of left distinct cosets of H in G , i.e. $k := [G:H]$. Since the cosets partition G and each coset has $|H|$ elements, it follows that $|G| = k \cdot |H|$. $\square$

proposition: $\gcd(a, n) = 1 \Rightarrow \text{ord}_n(a^k) = \frac{\text{ord}_n a}{\gcd(k, \text{ord}_n a)}$

Corollary: If there is a primitive root modulo n then there are exactly $\varphi(\varphi(n))$ primitive roots modulo n .

proof: Let r be a primitive root modulo n ($\text{ord } r = \varphi(n)$)

Then any other primitive root is from

$$\{1, r, r^2, \dots, r^{\varphi(n)-1}\}.$$

So any other primitive root is of the form r^k $0 \leq k \leq \varphi(n)$.

$$\Rightarrow \text{ord}(r^k) = \frac{\text{ord}(r)}{\gcd(k, \text{ord } r)} = \frac{\varphi(n)}{\gcd(k, \varphi(n))} = \varphi(n)$$

$\Rightarrow \gcd(k, \varphi(n)) = 1$. So there are exactly $\varphi(\varphi(n))$ such r^k .

example: $n=7$ $\varphi(7)=6 \rightsquigarrow 1, 2, 3, 6$

$$3^1=3, 3^2=2, 3^3=6 \Rightarrow 3^6=1$$

3 is a primitive root.

There are $\varphi(\varphi(7)) = \varphi(6) = 2$ primitive roots.

They are 3^k with $\gcd(k, \varphi(7)) = 1$:

$$3^1=3, 3^5=5$$

proposition: $\gcd(a, n) = 1$

$$\text{ord}(a^k) = \frac{\text{ord}_n a}{\gcd(k, \text{ord}_n a)}$$

proof: Set $d := \gcd(k, \text{ord}_n a)$. Observe $d|k, d|\text{ord}_n a$.

Observe $\frac{k}{d}, \frac{\text{ord}_n a}{d} \in \mathbb{N}$.

$$\text{Take } (a^k)^{\frac{\text{ord}_n a}{d}} = (a^{\text{ord}_n a})^{\frac{k}{d}} = 1^{\frac{k}{d}} \equiv 1 \pmod{n}.$$

$$\Rightarrow \text{ord}(a^k) \mid \frac{\text{ord}_n a}{d} \quad (1)$$

$$a^{k \cdot \text{ord}(a^k)} = (a^k)^{\text{ord}(a^k)} \equiv 1 \pmod{n}.$$

$$\Rightarrow \text{ord } a \mid k \cdot \text{ord}(a^k)$$

$$\Rightarrow \frac{\text{ord } a}{d} \mid \frac{k}{d} \cdot \text{ord}(a^k) \quad (2)$$

$$\Rightarrow \frac{\text{ord } a}{d} \mid \text{ord}(a^k) \quad (3)$$

$$\Rightarrow \text{ord}(a^k) = \frac{\text{ord } a}{d}$$

□

Definition: The order of a modulo n is the least $m \in \mathbb{N}$ such that

$$a^m \equiv 1 \pmod{n}$$

write $\text{ord}_n a = m$.

proposition: $a^k \equiv 1 \pmod{n} \Leftrightarrow \text{ord}_n a \mid k$.

proof: $\Rightarrow$: Suppose that $a^k \equiv 1 \pmod{n}$. Use the division algorithm with k and $\text{ord}_n a$:

$$k = (\text{ord}_n a) \cdot q + r \text{ with } 0 \leq r < \text{ord}_n a$$

$$a^k \equiv 1 \pmod{n} \equiv a^{(\text{ord}_n a) \cdot q + r} \equiv \underbrace{(a^{\text{ord}_n a})^q}_{=1} \cdot a^r \equiv a^r \pmod{n}$$

$\Rightarrow r=0$, cause $0 \leq r < \text{ord}_n a$

$\Rightarrow \text{ord}_n a \mid k$

$\Leftarrow$: Suppose $\text{ord}_n a \mid k \Rightarrow k = (\text{ord}_n a) \cdot b$ with $b \in \mathbb{N}$.

$$a^k \equiv a^{\text{ord}_n a \cdot b} \equiv (a^{\text{ord}_n a})^b \equiv 1^b \equiv 1 \pmod{n} \quad \square$$

Very Important Corollary

$$\text{ord}_n a \mid \varphi(n)$$

proof: By Euler's Theorem we know $a^{\varphi(n)} \equiv 1 \pmod{n}$.

Apply proposition $\Rightarrow \text{ord}_n a \mid \varphi(n)$